Lab Assignment 3: Advanced Univariate, Bivariate, and Multivariate Analysis

Objective: Perform exploratory data analysis (EDA) across 1, 2, and 3+ features using Plotnine (ggplot2 syntax in Python) and stacked visualizations.
Dataset: `diamonds` (53,940 rows, 10 columns; ships with plotnine).
Tasks: (1) Univariate: Plotnine histogram of price with chosen bin width and theme; (2) Bivariate: stacked (100%) bar plot of cut vs clarity; (3) Multivariate: faceted scatter plot (carat, price, clarity, cut) and pair plot (hue = cut).
Run with: `python lab3_advanced_eda_plotnine.py`

1. Python Code

```
# ======================================================================
# Lab Assignment 3: Advanced Univariate, Bivariate, and Multivariate
#                   Analysis
# ----------------------------------------------------------------------
# Objective : Exploratory data analysis (EDA) across 1, 2 and 3+
#             features using Plotnine (ggplot2 syntax in Python) and
#             stacked visualizations.
# Dataset   : "diamonds" (ships with plotnine: plotnine.data.diamonds)
# Run with  : python lab3_advanced_eda_plotnine.py
# Requires  : pip install plotnine pandas seaborn matplotlib
# Output    : Step-by-step console log + PNG plots in ./plots/
# ======================================================================

import os
import warnings
from datetime import datetime

import matplotlib
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import plotnine
import seaborn as sns
from plotnine import (aes, element_text, facet_wrap, geom_bar, geom_histogram,
                      geom_point, geom_vline, ggplot, labs, position_fill,
                      scale_color_cmap_d, scale_fill_brewer,
                      scale_y_continuous, theme, theme_bw, theme_minimal)
from plotnine.data import diamonds

warnings.filterwarnings("ignore")  # keep the console log clean


# ---------- Helper: print a clearly visible step header --------------
def step(n, title):
    print("\n" + "=" * 70)
    print(f"STEP {n} : {title}")
    print("=" * 70)


OUT_DIR = "plots"
os.makedirs(OUT_DIR, exist_ok=True)


def save_gg(plot, file, width=8, height=5):
    path = os.path.join(OUT_DIR, file)
    plot.save(path, width=width, height=height, dpi=110, verbose=False)
    print(">> Plot saved to:", path)


pd.set_option("display.width", 110)
print("Lab Assignment 3 - Advanced Univariate, Bivariate & Multivariate Analysis")
print("Python libraries : plotnine", plotnine.__version__, "| pandas", pd.__version__,
      "| seaborn", sns.__version__)
print("Run on           :", datetime.now().strftime("%Y-%m-%d %H:%M:%S"))


```

```
# ======================================================================
# STEP 1 : Load and inspect the dataset
# ======================================================================
step(1, "Load the 'diamonds' dataset")

df = diamonds.copy()
print(">> Dataset loaded successfully.")
print(">> Shape (rows x columns):", df.shape)

print("\n>> Column data types:")
print(df.dtypes)

print("\n>> First 5 rows:")
print(df.head())

print("\n>> Missing values in the whole dataset:", int(df.isnull().sum().sum()))

print("\n>> Summary of key continuous features:")
print(df[["carat", "depth", "table", "price"]].describe().round(2))

print("\n>> Levels of the categorical features (ordered worst -> best):")
for col in ["cut", "color", "clarity"]:
    print(f"   {col:<8}: {list(df[col].cat.categories)}")


# ======================================================================
# STEP 2 : Univariate analysis - Plotnine histogram of price
# ======================================================================
step(2, "Univariate: Plotnine Histogram of 'price' (bin width + theme)")

print(">> Effect of bin width on the number of bins:")
price_range = df["price"].max() - df["price"].min()
for bw in [250, 500, 1000]:
    print(f"   binwidth = {bw:>4}  ->  about {int(np.ceil(price_range / bw))} bins")

BINWIDTH = 500
print(f">> Chosen binwidth = {BINWIDTH} (fine enough to show shape, not too noisy)")
print(f">> Mean price   = ${df['price'].mean():,.2f}")
print(f">> Median price = ${df['price'].median():,.2f}")
print(f">> Skewness     = {df['price'].skew():.3f}  (> 0 means right-skewed)")

hist_plot = (
    ggplot(df, aes(x="price"))
    + geom_histogram(binwidth=BINWIDTH, fill="#2c7fb8", color="white", boundary=0)
    + geom_vline(xintercept=df["price"].mean(), color="red",
                 linetype="dashed", size=1)
    + geom_vline(xintercept=df["price"].median(), color="darkgreen",
                 linetype="dotted", size=1)
    + labs(title="Distribution of Diamond Prices",
           subtitle=f"binwidth = {BINWIDTH};  red dashed = mean,  green dotted = median",
           x="Price (USD)", y="Number of diamonds")
    + theme_minimal()
    + theme(plot_title=element_text(size=14, weight="bold"))
)
save_gg(hist_plot, "1_univariate_histogram_price.png")

print(">> Interpretation: price is strongly right-skewed - most diamonds cost",
      "under $5,000,\n   with a long tail of expensive stones up to $18,823.")


# ======================================================================
# STEP 3 : Bivariate analysis - Stacked bar plot of cut vs clarity
# ======================================================================
step(3, "Bivariate: Stacked Bar Plot of 'cut' vs 'clarity' (relative %)")

print(">> Counts (crosstab of cut x clarity):")
print(pd.crosstab(df["cut"], df["clarity"]))

print("\n>> Relative distribution - row percentages (each cut sums to 100%):")
print((pd.crosstab(df["cut"], df["clarity"], normalize="index") * 100).round(1))

stacked_plot = (
    ggplot(df, aes(x="cut", fill="clarity"))
    + geom_bar(position=position_fill(), color="white", size=0.2)
    + scale_y_continuous(labels=lambda v: [f"{x:.0%}" for x in v])
    + scale_fill_brewer(type="div", palette="RdYlBu")
```

```
    + labs(title="Relative Distribution of Clarity within each Cut",
           x="Cut quality", y="Share of diamonds", fill="Clarity")
    + theme_bw()
    + theme(plot_title=element_text(size=14, weight="bold"))
)
save_gg(stacked_plot, "2_bivariate_stacked_bar_cut_clarity.png")

ideal_if = (pd.crosstab(df["cut"], df["clarity"], normalize="index") * 100).loc["Ideal", "IF"]
fair_if = (pd.crosstab(df["cut"], df["clarity"], normalize="index") * 100).loc["Fair", "IF"]
print(">> Interpretation: better cuts have a larger share of high-clarity stones -\n"
      f"   e.g. IF clarity is {ideal_if:.1f}% of Ideal cuts but only {fair_if:.1f}% of Fair cuts.")


# ======================================================================
# STEP 4 : Multivariate analysis - faceted scatter (4 variables)
# ======================================================================
step(4, "Multivariate: Faceted Scatter - carat vs price, color = clarity, facet = cut")

sample = df[df["carat"] <= 3].sample(n=5000, random_state=42)
print(">> Using a random sample of 5,000 diamonds up to 3 carat (random_state=42)\n"
      "   to avoid overplotting.")
print(">> Variables encoded: x = carat, y = price, color = clarity, facet = cut")

print("\n>> Correlation of carat and price within each cut:")
print(df.groupby("cut", observed=True)
        .apply(lambda g: g["carat"].corr(g["price"]), include_groups=False)
        .round(3).rename("corr(carat, price)"))

print("\n>> Interaction effect - median price of ~1 carat diamonds (0.95-1.05 ct)")
print("   by cut and clarity (holding carat fixed shows the clarity effect):")
one_ct = df[df["carat"].between(0.95, 1.05)]
print(one_ct.pivot_table(values="price", index="cut", columns="clarity",
                         aggfunc="median", observed=True).round(0))

facet_plot = (
    ggplot(sample, aes(x="carat", y="price", color="clarity"))
    + geom_point(alpha=0.6, size=1.2)
    + facet_wrap("~cut", ncol=3)
    + scale_color_cmap_d("viridis")
    + labs(title="Carat vs Price by Clarity, faceted by Cut",
           x="Carat", y="Price (USD)", color="Clarity")
    + theme_bw()
    + theme(plot_title=element_text(size=14, weight="bold"))
)
save_gg(facet_plot, "3_multivariate_faceted_scatter.png", width=10, height=6.5)

print(">> Interpretation: price rises steeply with carat in every cut; within the\n"
      "   same carat, higher clarity (yellow/green) stones sit above lower clarity\n"
      "   (purple) ones, so carat and clarity interact to drive price in every cut.")


# ======================================================================
# STEP 5 : Multivariate analysis - pair plot with hue
# ======================================================================
step(5, "Multivariate: Pair Plot of carat, depth, table, price (hue = cut)")

pair_vars = ["carat", "depth", "table", "price"]
print(">> Correlation matrix of the pair-plot variables:")
print(df[pair_vars].corr().round(3))

small = df.sample(n=1500, random_state=1)
g = sns.pairplot(small, vars=pair_vars, hue="cut", palette="viridis",
                 plot_kws={"alpha": 0.5, "s": 12}, corner=True, height=2.1)
g.figure.suptitle("Pair Plot of Diamond Features (hue = cut, n = 1,500 sample)",
                  y=1.02, fontsize=13, fontweight="bold")
path = os.path.join(OUT_DIR, "4_multivariate_pairplot.png")
g.savefig(path, dpi=100, bbox_inches="tight")
plt.close(g.figure)
print(">> Plot saved to:", path)

print(">> Interpretation: carat and price are strongly correlated (r = "
      f"{df['carat'].corr(df['price']):.3f});\n"
      "   depth and table show little relation to price.")


# ======================================================================
```

```
# Done
# ======================================================================
step(6, "Lab Assignment 3 completed")
print("Files generated:")
for f in sorted(os.listdir(OUT_DIR)):
    print("  -", os.path.join(OUT_DIR, f))
```

2. Execution Output (console printout)

```
Lab Assignment 3 - Advanced Univariate, Bivariate & Multivariate Analysis
Python libraries : plotnine 0.15.8 | pandas 3.0.5 | seaborn 0.13.2
Run on           : 2026-10-26 21:30:15

======================================================================
STEP 1 : Load the 'diamonds' dataset
======================================================================
>> Dataset loaded successfully.
>> Shape (rows x columns): (53940, 10)

>> Column data types:
carat       float64
cut        category
color      category
clarity    category
depth       float64
table       float64
price         int64
x           float64
y           float64
z           float64
dtype: object

>> First 5 rows:
   carat      cut color clarity  depth  table  price     x     y     z
0   0.23    Ideal     E     SI2   61.5   55.0    326  3.95  3.98  2.43
1   0.21  Premium     E     SI1   59.8   61.0    326  3.89  3.84  2.31
2   0.23     Good     E     VS1   56.9   65.0    327  4.05  4.07  2.31
3   0.29  Premium     I     VS2   62.4   58.0    334  4.20  4.23  2.63
4   0.31     Good     J     SI2   63.3   58.0    335  4.34  4.35  2.75

>> Missing values in the whole dataset: 0
>> Summary of key continuous features:
          carat     depth     table     price
count  53940.00  53940.00  53940.00  53940.00
mean       0.80     61.75     57.46   3932.80
std        0.47      1.43      2.23   3989.44
min        0.20     43.00     43.00    326.00
25%        0.40     61.00     56.00    950.00
50%        0.70     61.80     57.00   2401.00
75%        1.04     62.50     59.00   5324.25
max        5.01     79.00     95.00  18823.00

>> Levels of the categorical features (ordered worst -> best):
   cut     : ['Fair', 'Good', 'Very Good', 'Premium', 'Ideal']
   color   : ['D', 'E', 'F', 'G', 'H', 'I', 'J']
   clarity : ['I1', 'SI2', 'SI1', 'VS2', 'VS1', 'VVS2', 'VVS1', 'IF']
======================================================================
STEP 2 : Univariate: Plotnine Histogram of 'price' (bin width + theme)
======================================================================
>> Effect of bin width on the number of bins:
   binwidth =  250  ->  about 74 bins
   binwidth =  500  ->  about 37 bins
   binwidth = 1000  ->  about 19 bins
>> Chosen binwidth = 500 (fine enough to show shape, not too noisy)
>> Mean price   = $3,932.80
>> Median price = $2,401.00
>> Skewness     = 1.618  (> 0 means right-skewed)
>> Plot saved to: plots/1_univariate_histogram_price.png
>> Interpretation: price is strongly right-skewed - most diamonds cost under $5,000,
   with a long tail of expensive stones up to $18,823.
======================================================================
STEP 3 : Bivariate: Stacked Bar Plot of 'cut' vs 'clarity' (relative %)
======================================================================
>> Counts (crosstab of cut x clarity):
clarity     I1   SI2   SI1   VS2   VS1  VVS2  VVS1    IF
cut
Fair       210   466   408   261   170    69    17     9
Good        96  1081  1560   978   648   286   186    71
Very Good   84  2100  3240  2591  1775  1235   789   268
Premium    205  2949  3575  3357  1989   870   616   230
Ideal      146  2598  4282  5071  3589  2606  2047  1212

>> Relative distribution - row percentages (each cut sums to 100%):
```

```
clarity      I1   SI2   SI1   VS2   VS1  VVS2  VVS1   IF
cut
Fair       13.0  28.9  25.3  16.2  10.6   4.3   1.1  0.6
Good        2.0  22.0  31.8  19.9  13.2   5.8   3.8  1.4
Very Good   0.7  17.4  26.8  21.4  14.7  10.2   6.5  2.2
Premium     1.5  21.4  25.9  24.3  14.4   6.3   4.5  1.7
Ideal       0.7  12.1  19.9  23.5  16.7  12.1   9.5  5.6
>> Plot saved to: plots/2_bivariate_stacked_bar_cut_clarity.png
>> Interpretation: better cuts have a larger share of high-clarity stones -
   e.g. IF clarity is 5.6% of Ideal cuts but only 0.6% of Fair cuts.
======================================================================
STEP 4 : Multivariate: Faceted Scatter - carat vs price, color = clarity, facet = cut
======================================================================
>> Using a random sample of 5,000 diamonds up to 3 carat (random_state=42)
   to avoid overplotting.
>> Variables encoded: x = carat, y = price, color = clarity, facet = cut

>> Correlation of carat and price within each cut:
cut
Fair         0.859
Good         0.922
Very Good    0.926
Premium      0.925
Ideal        0.931
Name: corr(carat, price), dtype: float64
>> Interaction effect - median price of ~1 carat diamonds (0.95-1.05 ct)
   by cut and clarity (holding carat fixed shows the clarity effect):
clarity        I1     SI2     SI1     VS2     VS1    VVS2    VVS1       IF
cut
Fair       1954.0  3466.0  4256.0  4887.0  4984.0  5832.0  8529.0      NaN
Good       2798.0  4002.0  4679.0  5744.0  6354.0  7442.0  7373.0   9867.0
Very Good  3070.0  4260.0  4932.0  6159.0  6785.0  8265.0  9222.0  10305.0
Premium    2815.0  4155.0  4899.0  6053.0  6730.0  8470.0  9550.0   9012.0
Ideal      3276.0  4441.0  5103.0  6432.0  7240.0  9010.0  8938.0  10787.0
>> Plot saved to: plots/3_multivariate_faceted_scatter.png
>> Interpretation: price rises steeply with carat in every cut; within the
   same carat, higher clarity (yellow/green) stones sit above lower clarity
   (purple) ones, so carat and clarity interact to drive price in every cut.

======================================================================
STEP 5 : Multivariate: Pair Plot of carat, depth, table, price (hue = cut)
======================================================================
>> Correlation matrix of the pair-plot variables:
       carat  depth  table  price
carat  1.000  0.028  0.182  0.922
depth  0.028  1.000 -0.296 -0.011
table  0.182 -0.296  1.000  0.127
price  0.922 -0.011  0.127  1.000
>> Plot saved to: plots/4_multivariate_pairplot.png
>> Interpretation: carat and price are strongly correlated (r = 0.922);
   depth and table show little relation to price.

======================================================================
STEP 6 : Lab Assignment 3 completed
======================================================================
Files generated:
  - plots/1_univariate_histogram_price.png
  - plots/2_bivariate_stacked_bar_cut_clarity.png
  - plots/3_multivariate_faceted_scatter.png
  - plots/4_multivariate_pairplot.png
```

3. Output Plots

Figure 1: Univariate - Histogram of Price (binwidth = 500, theme_minimal)

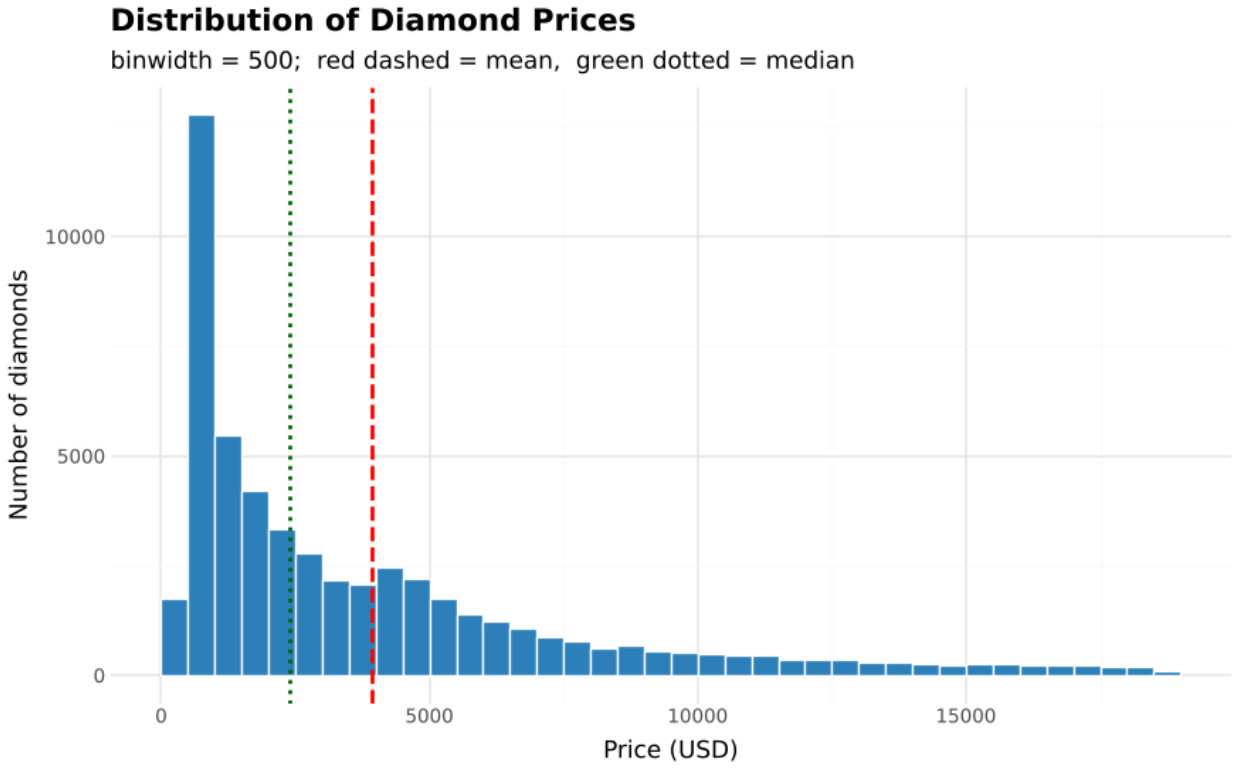


Figure 2: Bivariate - Stacked Bar Plot of Clarity within each Cut

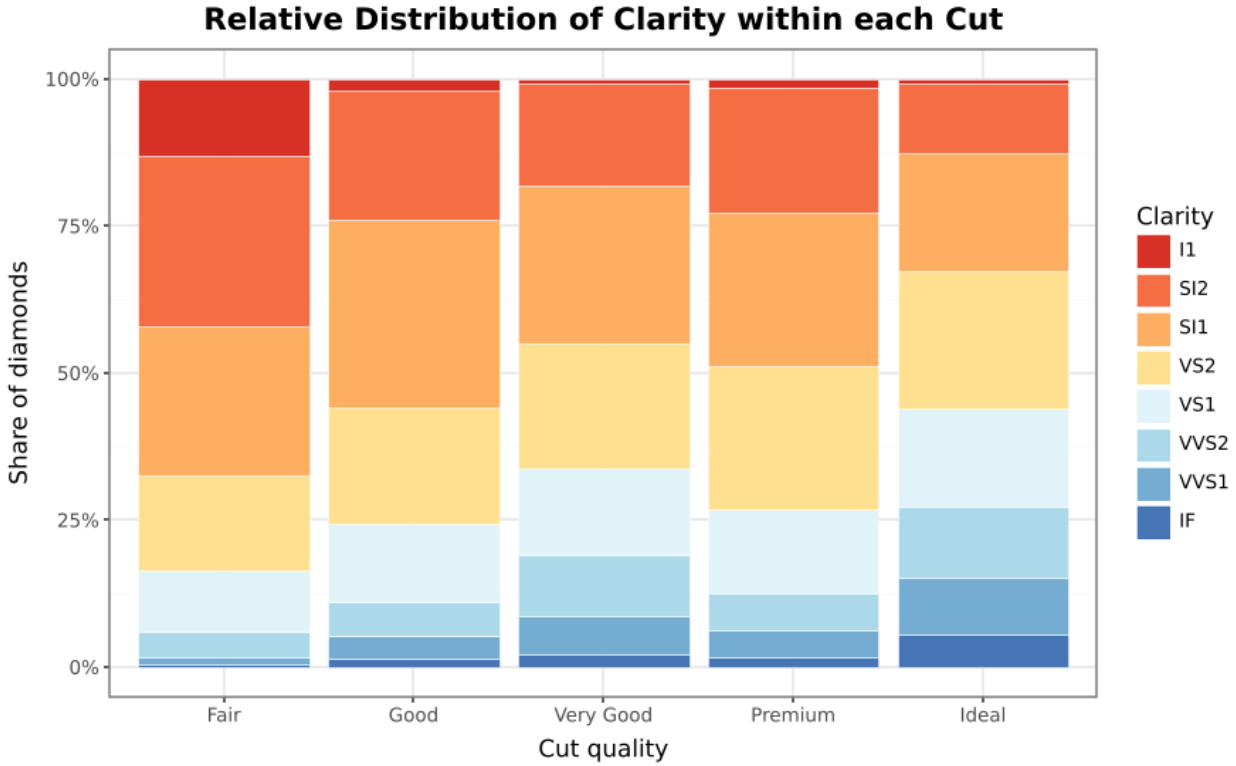


Figure 3: Multivariate - Carat vs Price, color = Clarity, facet = Cut



Figure 4: Multivariate - Pair Plot of carat, depth, table, price (hue = cut)

